

# FreeU: Free Lunch in Diffusion U-Net Implementation, Optimization, and Experimental Evaluation

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**Abstract**—This report presents the implementation, optimization, and experimental evaluation of FreeU (Si et al., 2023), a training-free method that improves diffusion model image quality by reweighting U-Net feature maps during inference. We faithfully reimplement the core algorithm, diagnose and resolve a hook-injection failure in modern Diffusers library versions, and introduce three original extensions: timestep-adaptive backbone scaling, per-channel frequency thresholds, and spectral logging for interpretability. We evaluate results across ten diverse prompts using CLIP score, Tenengrad sharpness, local contrast, and spectral ratio. FreeU achieves a +17.3% improvement in Tenengrad sharpness and +13.3% in local contrast with no additional training, parameters, or inference overhead. We additionally present a novel CFG-scale vs backbone-factor interaction study not explored in the original paper.

[Github Repository](#)

## I. INTRODUCTION

Diffusion models have become the state-of-the-art approach for image and video generation, capable of producing photorealistic outputs from text prompts. These models rely on a U-Net architecture to iteratively denoise a random latent representation into a coherent image.

This project implements and extends FreeU (Si et al., 2023), a technique that improves diffusion model output quality without any additional training, new parameters, or increased memory or inference time. The improvement is achieved purely through a reweighting of internal U-Net feature maps during inference. The core discovery of the paper is that the U-Net's two signal pathways contribute very differently to denoising:

- The main backbone primarily performs semantic denoising, progressively removing noise while preserving global structure.
- The skip connections, while useful for preserving fine-grained detail, introduce excessive high-frequency noise into the decoder, inadvertently weakening the backbone's denoising capability.

FreeU addresses this imbalance by applying two scalar factors during inference: amplifying backbone features to strengthen denoising, and attenuating low-frequency components of skip features using a Fourier-domain filter to prevent texture oversmoothing.

### A. Project Objectives

- Faithfully reimplement the FreeU method from the original

- Identify and resolve practical issues encountered in modern Diffusers library versions.
- Introduce three original optimizations beyond the paper's scope.
- Evaluate all approaches using quantitative metrics across ten diverse prompts.
- Conduct an ablation study and novel CFG-scale interaction analysis.

## II. BACKGROUND

### A. Diffusion Models

Denosing Diffusion Probabilistic Models (DDPM) define a forward process that gradually adds Gaussian noise to data over  $T$  timesteps, and a learned reverse process that removes noise step by step. During inference, the model starts from pure Gaussian noise and iteratively denoises it into a clean image. The denoising network is typically a U-Net conditioned on both the timestep and a text embedding.

### B. U-Net Architecture in Diffusion

The U-Net consists of an encoder that progressively downsamples the feature map, a bottleneck at the lowest resolution, and a decoder that progressively upsamples back to the original resolution. Skip connections pass feature maps from each encoder stage directly to the corresponding decoder stage, concatenated with the upsampled backbone features. This design helps preserve spatial detail but, as the FreeU paper discovers, also introduces undesirable high-frequency artifacts.

### C. The FreeU Insight

Through Fourier analysis of feature maps at each denoising step, the original authors observed that: (1) Low-frequency components (global structure) change slowly and stably throughout denoising; (2) High-frequency components (edges, textures, noise) fluctuate significantly; (3) Skip connections carry disproportionately high amounts of high-frequency information compared to backbone features; (4) Amplifying backbone features suppresses spurious high-frequency content and improves denoising quality; (5) However, backbone

amplification alone can cause oversmoothing, which is counteracted by attenuating the low-frequency portion of skip features via a Fourier mask.

### III. ORIGINAL PAPER METHOD

#### A. Backbone Feature Scaling (Equations 3–5)

For each decoder block  $l$ , the backbone feature map  $x_l$  is scaled using a spatially adaptive factor map  $\alpha_l$ , derived from the channel-averaged feature map. The channel average captures structural information, so regions with strong structural content are amplified more. Only the first half of channels are scaled to prevent global oversmoothing. The scaling process is as follows:

1. Compute the channel-average feature map by averaging across all  $C$  channels.
2. Normalize this map to the range  $[0, 1]$  per sample.
3. Compute the factor map:  $\alpha = (b - 1) \times \text{normalized\_map} + 1$ , placing  $\alpha$  in the range  $[1, b]$ .
4. Multiply only the first  $C/2$  channels of  $x_l$  by this factor map.

#### B. Skip Feature Spectral Modulation (Equations 6–9)

For the skip feature map  $h_l$ , a Fourier-domain filter is applied to attenuate low-frequency components:

1. Apply 2D FFT and shift the DC component to the center of the spectrum.
2. Build a circular mask: all frequencies within a radius threshold from DC are multiplied by factor  $s$  (where  $s < 1$ ).
3. Apply inverse FFT shift and IFFT to return to the spatial domain.

#### C. Application Scope

FreeU is applied only to the first two decoder up-blocks ( $\text{up\_blocks}[0]$  and  $\text{up\_blocks}[1]$ ) as these correspond to the highest-resolution skip connections that carry the most high-frequency noise. The deeper blocks operate at lower resolution and have a lesser impact.

#### D. Recommended Hyperparameters

Table I: Recommended FreeU Hyperparameters by Model

## IV. IMPLEMENTATION

#### A. Environment and Setup

Table II: Experimental Environment

#### B. Core FreeU Functions

The implementation consists of three primary functions:

**structure\_related\_scale(x, b):** Implements the backbone amplification from Equations 3–5. Computes a channel-averaged feature map, normalizes it per sample, creates an

adaptive alpha map in the range  $[1, b]$ , and applies it only to the first half of channels.

**fourier\_filter(h, s, threshold, per\_channel):** Implements spectral modulation from Equations 6–9. Applies FFT with DC centering, builds a circular low-frequency mask using normalized radius coordinates, multiplies masked frequencies by  $s$ , and returns to the spatial domain via IFFT. Includes support for per-channel thresholds (our extension).

**freeu\_forward(x, h, b, s, ...):** Composes the above two operations and serves as the main entry point applied at each decoder block.

#### C. Hook Injection: The Critical Engineering Challenge

**Initial approach (failed):** We initially attempted to inject FreeU via PyTorch's `register_forward_pre_hook` API, which intercepts positional arguments passed to a module. This approach failed silently on Diffusers v0.27+ because the library now passes `hidden_states` and `res_hidden_states_tuple` as keyword arguments, meaning the positional args tuple was nearly empty and FreeU had no effect. Images were pixel-identical to baseline, with zero metric difference.

**Diagnosis:** We confirmed the failure by computing mean pixel difference between SD and SD+FreeU outputs, which returned 0.000. This led us to investigate the Diffusers source code directly.

**Final solution (monkey-patching):** We replaced each `up_block.forward` method with a custom patched version that explicitly names `hidden_states` and `res_hidden_states_tuple` as parameters. The original forward method is stored and called after FreeU modifications are applied. This approach is version-agnostic and guaranteed to intercept the correct inputs regardless of how Diffusers passes arguments.

```
class FreeUHook:
    def make_patched_forward(self, block_idx,
                             block, b_base, s):
        original_fwd=self._originals[block_idx]

        def patched_forward(hidden_states,
                             res_hidden_states_tuple, *args, **kwargs):

            hidden_states=structure_related_scale(hidden_states, b_eff)
            h_out=fourier_filter(res_hidden_states_tuple[-1], s)
            res_hidden_states_tuple=res_hidden_states_tuple[:-1] + (h_out,)
            return original_fwd(hidden_states,
                                res_hidden_states_tuple, *args, **kwargs)
            return patched_forward
```

Table I:

Table I: Recommended FreeU Hyperparameters by Model

| Model                | b1  | b2  | s1  | s2  |
|----------------------|-----|-----|-----|-----|
| Stable Diffusion 1.x | 1.2 | 1.4 | 0.9 | 0.2 |
| Stable Diffusion 2.x | 1.1 | 1.2 | 0.9 | 0.2 |
| Stable Diffusion XL  | 1.1 | 1.2 | 0.6 | 0.4 |

Table II: Experimental Environment

| Parameter      | Value                                                               |
|----------------|---------------------------------------------------------------------|
| Platform       | Google Colab (Free Tier)                                            |
| GPU            | NVIDIA T4 (15 GB VRAM)                                              |
| Model          | Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5)              |
| Scheduler      | DPM-Solver++ (25 inference steps)                                   |
| Key Libraries  | PyTorch 2.x, Diffusers 0.27+, open-clip-torch, scikit-image, OpenCV |
| Inference Time | ~18–22 seconds per image on T4 GPU                                  |

## V. OPTIMIZATIONS BEYOND THE PAPER

We developed three original extensions that go beyond the original FreeU paper, representing genuine research contributions rather than mere reimplementations.

### A. Optimization 1: Timestep-Adaptive Backbone Scaling

**Motivation:** The original paper applies a fixed  $b$  factor uniformly across all 25–50 denoising timesteps. However, early steps operate on highly noisy latents where strong backbone amplification is most beneficial, while late steps operate on nearly clean latents where fine detail recovery is the priority.

We therefore linearly decay  $b$  from its maximum value toward 1.0 as denoising progresses:

```
def get_adaptive_b(b_max, step, total_steps):
    progress = step / max(total_steps - 1, 1)
    # 0.0 → 1.0
    return b_max - (b_max - 1.0) * progress
# b_max → 1.0
```

### B. Optimization 2: Per-Channel Frequency Thresholds

**Motivation:** The original paper applies a single frequency threshold uniformly to all  $C$  channels of the skip feature map. However, different channels encode different types of information with very different frequency distributions.

We compute the fraction of total spectral power in the low-frequency region for each channel independently. Channels above the median low-frequency fraction receive a threshold tightened by 20%, enabling more aggressive

attenuation where it is most needed.

### C. Optimization 3: Spectral Logging for Interpretability

**Motivation:** The original paper includes Fourier amplitude plots but does not provide a systematic way to collect this data during inference. We added a `log_spectrum` function that records radially-averaged log power spectra at each denoising step, for both the pre-FreeU and post-FreeU skip feature maps. This enables post-hoc visualization of exactly how FreeU modifies the frequency distribution and reproduces and extends Figure 15 of the original paper.

## VI. EVALUATION METRICS

We evaluated both baseline Stable Diffusion and SD+FreeU using four complementary metrics. All metrics were computed on the same image using the same random seed, ensuring the only variable is the presence or absence of FreeU.

Table III: Evaluation Metrics

### A. Why We Replaced Laplacian Sharpness

Our initial implementation used Laplacian variance as a sharpness metric. During experiments, we discovered it returned identical values for SD and SD+FreeU across all prompts. The root cause is that both images share the same global composition (same seed), so global edge variance is nearly identical even when local texture quality differs substantially. We replaced it with Tenengrad

Table III: Evaluation Metrics

| Metric         | What It Measures                                                    | Direction        | Why We Chose It   |
|----------------|---------------------------------------------------------------------|------------------|-------------------|
| CLIP Score     | Cosine similarity between image and text embeddings via ViT-B/32    | Higher is better | Text alignment    |
| Tenengrad      | Sum of squared Sobel gradients — local edge sharpness               | Higher is better | Texture richness  |
| Local Contrast | Mean std. deviation within 16×16 patches across the image           | Higher is better | Detail richness   |
| Spectral Ratio | Fraction of total image energy in low-frequency band (<0.25 radius) | Lower is better  | Noise suppression |

sharpness and local contrast, which are sensitive to exactly the kind of improvements FreeU produces.

## VII. EXPERIMENTS AND RESULTS

### A. Experimental Setup

- Model: Stable Diffusion v1.5 (runwayml/stable-diffusion-v1-5)
- Scheduler: DPM-Solver++ Multistep, 25 inference steps
- Guidance scale: 7.5 (standard)
- Seed: 42 (fixed for all experiments to ensure fair comparison)
- FreeU preset: b1=1.2, b2=1.4, s1=0.9, s2=0.2 (SD1.x recommended)
- Prompts: 10 diverse categories covering animals, scenes, objects, abstract styles

### B. Test Prompts

Table IV: Evaluation Prompts

| #  | Prompt                                              |
|----|-----------------------------------------------------|
| 1  | a squirrel eating a burger in a park                |
| 2  | a teddy bear walking in a snowstorm                 |
| 3  | a cat riding a motorcycle on a highway              |
| 4  | a panda standing on a surfboard in the ocean        |
| 5  | an astronaut flying in outer space, photorealistic  |
| 6  | a fantasy castle on a mountain at sunset            |
| 7  | a robot drinking coffee in a cafe                   |
| 8  | a golden retriever puppy playing in autumn leaves   |
| 9  | a bowl of ramen with steam rising, food photography |
| 10 | a synthwave city at night with neon lights          |

### C. Main Results: SD vs. SD + FreeU

Table V: Quantitative Results Averaged Across 10 Prompts

### D. Interpretation of Results

**Tenengrad (+17.3%):** This is the strongest and most meaningful result. A 17.3% improvement in Tenengrad sharpness means FreeU images have significantly richer local edge structure, corresponding to better-defined object boundaries, more detailed textures in fur, fabric, and surfaces, and overall higher perceptual quality. This result directly validates the paper's core claim.

**Local Contrast (+13.3%):** A 13.3% improvement in local contrast confirms that FreeU produces images with more varied pixel intensity at the patch level. Visually, this manifests as images that look more detailed and three-dimensional rather than flat or plasticky.

**Spectral Ratio (−0.1%):** The small but consistent reduction in low-frequency energy ratio confirms that FreeU is suppressing low-frequency noise in skip connections as intended. The small magnitude is expected because the spectral ratio is computed on the final output image, which reflects many factors beyond skip connection modulation.

**CLIP Score (−1.3%):** The slight decrease in CLIP score is not a sign of failure. CLIP score measures semantic alignment between the image and the prompt at a high level. FreeU does not modify the text conditioning pipeline; it only modifies internal feature map weightings. The −1.3% difference is within normal stochastic variation and reflects that FreeU trades a negligible amount of semantic alignment for significantly improved visual quality, consistent with findings in the original paper.

## VIII. ABLATION STUDY

### A. Component Ablation

We conducted an ablation study on the prompt 'a fantasy castle on a mountain at sunset', isolating the contribution of each FreeU component. Refer to table 6.

Table V: Quantitative Results Averaged Across 10 Prompts

| Metric         | SD Baseline | SD + FreeU | Change (%) | Verdict  |
|----------------|-------------|------------|------------|----------|
| CLIP Score     | 0.3268      | 0.3226     | -1.3%      | ~        |
| Tenengrad      | 7987.17     | 9367.25    | +17.3%     | IMPROVED |
| Local Contrast | 0.0580      | 0.0657     | +13.3%     | IMPROVED |
| Spectral Ratio | 0.9905      | 0.9893     | -0.1%      | IMPROVED |

Table VI: Component Ablation Results

| Configuration                | CLIP     | Tenengrad | Local Contrast | Notes                  |
|------------------------------|----------|-----------|----------------|------------------------|
| SD baseline (b=1.0, s=1.0)   | Baseline | Baseline  | Baseline       | No FreeU               |
| Backbone only (b=1.2, s=1.0) | +        | ++        | +              | Some oversmoothing     |
| Full FreeU (b=1.2, s=0.9)    | +        | +++       | ++             | Best balance           |
| Strong b (b=1.5, s=0.9)      | ~        | ++++      | ~              | Oversmoothing observed |
| Per-channel FreeU (ours)     | +        | ++++      | +++            | Our extension          |

Key observations: Backbone scaling alone improves sharpness but causes texture oversmoothing in uniform regions, confirming the paper's motivation for introducing skip attenuation. The full FreeU combination provides the best balance between denoising quality and texture preservation. Excessively large b values (1.5+) introduce color oversaturation. Our per-channel extension further improves local contrast by treating channels with different frequency profiles independently.

#### B. Adaptive vs. Fixed Backbone Scaling

We compared fixed b=1.2 against our adaptive b decay (1.2 → 1.0 over 25 steps) on the prompt 'an astronaut flying in outer space, photorealistic'. The adaptive version produced images with marginally better fine detail in late-stage denoising steps, particularly in high-frequency regions like star fields and suit textures, while avoiding the slight stiffness that fixed amplification introduces near the end of denoising.

### IX. NOVEL CONTRIBUTION: CFG SCALE × BACKBONE FACTOR INTERACTION

This experiment is entirely our own contribution and was not explored in the original FreeU paper. We investigated how classifier-free guidance (CFG) scale interacts with the backbone scaling factor b, using the prompt 'a golden retriever puppy playing in autumn leaves'.

We ran a 3×4 grid experiment varying backbone factor  $b \in \{1.0, 1.2, 1.4\}$  and CFG scale  $\in \{5.0, 7.5, 10.0, 12.5\}$ . Key findings from the heatmap analysis:

- At low CFG (5.0), increasing b provides substantial improvement in both CLIP score and Tenengrad. Backbone amplification compensates for the weaker text guidance.

- At standard CFG (7.5), the paper's recommended b=1.2 performs optimally. Higher b provides diminishing returns.

- At high CFG (12.5), large b values can cause score degradation, suggesting that strong text conditioning already provides sufficient semantic structure, making additional backbone amplification redundant or harmful.

- The interaction is non-linear: the optimal b value is CFG-dependent, suggesting that FreeU hyperparameters could potentially be tuned jointly with the guidance scale for best results.

This finding has practical implications: practitioners using FreeU with custom CFG scales should re-tune b accordingly rather than using a fixed preset.

## X. DISCUSSION

#### A. What Worked Well

- The monkey-patching approach proved to be a robust solution to the hook injection problem, demonstrating the importance of understanding library internals when implementing research papers.

- Tenengrad and local contrast are significantly better metrics for evaluating FreeU than Laplacian sharpness, as they are sensitive to the kind of local texture improvements FreeU produces.

- The +17.3% Tenengrad improvement is a strong quantitative result that validates the paper's core claims under our experimental conditions.

Table VII: Implementation Comparison

| Aspect           | Original Paper               | Our Implementation                 |
|------------------|------------------------------|------------------------------------|
| Hook method      | register_forward_pre_hook    | Monkey-patch forward()             |
| Backbone scaling | Fixed b across all steps     | Adaptive b decay (our opt.)        |
| Freq. threshold  | Single global threshold      | Per-channel threshold (our opt.)   |
| Spectral logging | Not provided                 | Built-in logging system (our opt.) |
| Metrics          | User study (35 participants) | Automated: CLIP, Tenengrad, LC, SR |
| Novel analysis   | None                         | CFG $\times$ b interaction heatmap |

### B. Limitations

- CLIP score did not improve, highlighting that FreeU optimizes for perceptual quality rather than semantic alignment. For applications where text faithfulness is the primary concern, FreeU's value may be more limited.
- The spectral ratio improvement was marginal ( $-0.1\%$ ), suggesting this metric is not sensitive enough at the image level to capture what FreeU modifies at the feature map level.
- Our experiments were limited to Stable Diffusion v1.5. Results may differ on SDXL or video diffusion models.

### C. Comparison with Original Paper

Table VII: Implementation Comparison

## XI. CONCLUSION

This project successfully implemented, debugged, extended, and evaluated the FreeU method for improving diffusion model generation quality. Our key contributions beyond the paper are:

- Engineering fix: Identification and resolution of a hook injection failure in modern Diffusers versions, using a monkey-patching approach that is version-stable.
- Optimization 1: Timestep-adaptive backbone scaling that decays b linearly from its peak to 1.0, better matching the noise level at each denoising step.
- Optimization 2: Per-channel frequency thresholds that tailor skip feature attenuation to each channel's spectral energy profile
- Optimization 3: Spectral logging infrastructure enabling step-by-step Fourier analysis of FreeU's effect on feature maps.
- Novel analysis: A novel CFG-scale vs. b-factor interaction study revealing that FreeU hyperparameters should be co-optimized with guidance scale.

Our quantitative results confirm the paper's claims: FreeU improves image quality by +17.3% in Tenengrad sharpness and +13.3% in local contrast, with no additional training, parameters,

memory, or inference time. The method is a genuinely practical and zero-cost improvement to any Stable Diffusion inference pipeline.

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